Bayesian Optimization of Multi-Material 3D-Printed Tensegrity-Inspired Structures for Energy Absorption

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Tensegrity-inspired architectures combine rigid compression members with a continuous network of soft tension elements, yielding lightweight assemblies with tunable nonlinear responses that are attractive for energy-absorbing applications such as protective equipment, packaging, and aerospace landing structures. Multi-material fused-deposition modeling (FDM) of polylactic acid (PLA) struts and thermoplastic polyurethane (TPU) tension elements makes it possible to fabricate diverse tensegrity-inspired geometries rapidly enough to support a high-throughput experimental search, while the rate-dependent damping of TPU is itself an asset for impact mitigation. We report an experiment-driven Bayesian optimization (BO) framework in which a Gaussian-process surrogate is trained directly on physical quasi-static compression and drop-weight impact data and used to recommend the next batch of designs to print and test. The framework parameterizes a family of tensegrity-inspired unit cells over strut geometry, tension-element cross-section, connectivity topology, and unit-cell tiling, and optimizes peak transmitted force, specific energy absorption, and compaction efficiency.

Keywords: tensegrity, multi-material 3D printing, Bayesian optimization, energy absorption, design of experiments

1 Introduction

Tensegrity structures are assemblies of rigid compression members suspended within a continuous network of tension members; their defining property—no two rigid bars touch—yields lightweight assemblies with remarkable strength-to-weight ratios and tunable nonlinear mechanical responses [1,2]. These properties have made tensegrity attractive across scales, from metamaterial unit cells [3–5] to compliant impact-absorbing robots developed for planetary landing [6–10]. Recent work has shown that *tensegrity-inspired* unit cells can be directly 3D-printed and exhibit load-limiting plateaus under drop-weight impact [11], opening a path toward additively manufactured architected absorbers.

Multi-material fused-deposition modeling (FDM) using polylactic acid (PLA) for rigid struts and thermoplastic polyurethane (TPU) for flexible tension elements enables rapid fabrication of diverse tensegrity-inspired geometries on a single platform [12,13]. Although extruded TPU does not behave as an idealized inextensible cable, its rate-dependent damping is a desirable property for impact-absorption applications and complements the elastic stiffness of the PLA compression members rather than competing with it.

Designing such architected absorbers is fundamentally a *design-of-experiments* problem: the configuration space spanned by strut geometry, tension-element cross-section, connectivity, and unit-cell tiling is large; each candidate must be physically printed and tested; and the relevant performance metrics—peak transmitted force, specific energy absorption (SEA), and compaction efficiency—are noisy and partially correlated. Bayesian optimization (BO) is well-matched to this regime, having driven breakthroughs from hyperparameter tuning of AlphaGo [14] to sustainable concrete [15], autonomous discovery of organic laser emitters [16], and architected materials [17–20].

1.0.1 Contributions.. This paper makes the following contributions:

(1) A parameterized family of multi-material 3D-printable tensegrity-inspired unit cells with PLA compression members and TPU tension elements wrapped using the core-encapsulation strategy of ~~**(author?)** [12]~~ Ye et al. [12] to ensure cyclic interface durability.
(2) An experiment-driven BO loop, built on BoTorch/Ax [21], that operates directly on physical quasi-static compression and drop-weight impact measurements, recommending the next designs to fabricate without requiring an accurate forward simulator.
(3) A two-fidelity escalation path that pairs the direct-experimental loop above with a planned higher-fidelity rung based on *pretensioned* tensegrity assemblies (true cables and measured pretension) for selected Pareto-optimal designs; the specific surrogate / fusion strategy is left open.

The remainder of the paper is organized as follows. Section 2 reviews tensegrity mechanics, multi-material 3D printing for energy absorption, and Bayesian optimization with an emphasis on multi-objective and noise-robust acquisition functions. Section 3 introduces the design parameterization, fabrication workflow, experimental protocols, and BO loop. Section 4 reports the results of the closed-loop experimental campaign, and Section 5 discusses the implications and limitations of the approach. Section 6 concludes.

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Figure placeholder: Closed-loop, experiment-driven design framework: parameterized PLA/TPU tensegrity-inspired unit cells are printed, tested under quasi-static compression and drop-weight impact, and the resulting performance data drive a Gaussian-process surrogate that recommends the next batch of designs.

Fig. 1 ***Placeholder.*** **Closed-loop, experiment-driven design framework: parameterized PLA/TPU tensegrity-inspired unit cells are printed, tested under quasi-static compression and drop-weight impact, and the resulting performance data drive a Gaussian-process surrogate that recommends the next batch of designs.**

2 Background

2.1 Tensegrity-Inspired Architectures for Energy Absorption. Tensegrity prisms and their planar extensions exhibit softening–stiffening response under compression [3–5], characteristics desirable for impact mitigation because the structure can absorb a large fraction of the input energy at a controlled, sub-injurious force plateau. ~~**(author?)** [22]~~ Santos [22] recently characterized a tensegrity energy-dissipation metamaterial reporting equivalent viscous damping on the order of ~23% with negligible residual strain per impact, underscoring the suitability of tensegrity-inspired architectures for repeated-impact use cases. The space program literature documents large-scale tensegrity-inspired absorbers for planetary landing [6–10] and a long history of crushable energy-absorbing landing systems [23–25]. ~~**(author?)** [11]~~ Pajunen et al. [11] demonstrated that truncated-tetrahedral tensegrity-inspired unit cells can be directly 3D-printed and exhibit load-limiting plateaus, motivating the present focus on additively manufactured TPU/PLA composites.

2.2 Multi-Material 3D Printing of PLA/TPU Composites. Multi-material rigid–soft FDM [12,13] enables PLA–TPU combinations on a single platform with tunable energy absorption. Crucially, ~~**(author?)** [12]~~ Ye et al. [12] introduced a *wrapping-based* strategy in which rigid cores are encapsulated by continuous soft skins, preventing interface delamination and enabling cyclic durability—a critical enabler for our fabrication approach. Recent multi-material PLA/TPU sandwich and layered composites [26,27] report quantitative bounds on stiffness, strength, and energy absorption for the same PLA–TPU pair used here, while FFF-PLA fatigue data [28,29] inform conservative design bounds for the rigid struts. Architected multi-material lattices with co-printed compliant skins have been explicitly designed for high specific energy absorption [30], ~~and~~ with FFF-specific multimaterial interface adhesion (notably TPU bonded to rigid co-printed substrates) characterized in [31]; recent surveys catalogue the rapidly expanding additively manufactured polymeric energy-absorption literature [32,33].

2.3 Bayesian Optimization for Architected-Material Design. Bayesian optimization is a sequential, model-based strategy for optimizing expensive black-box functions using a Gaussian-process surrogate [34,35]. Modern implementations such as BoTorch [21] support parallel batched evaluations and a wide variety of acquisition functions; recent algorithmic advances include log-EI for numerical robustness [36], noisy expected hypervolume improvement for multi-objective settings [37], input-noise-robust acquisitions [38], and evolution-guided constrained multi-objective BO for self-driving labs [39]. Mixed/categorical search spaces—needed here for the discrete connectivity-topology variable—have dedicated treatments in BOCS [40] and the one-hot-with-rounded-kernel construction of [41]. BO has been applied to mechanical metamaterial and structural design [17–19] and to additive manufacturing [20]. ~~**(author?)** [17]~~ Mo et al. [17] in particular established a multifidelity BO framework for architected materials by fusing low-fidelity simulations with high-fidelity ground truth via nonlinear information-fusion priors [42]; the present work shifts the emphasis from simulation surrogates to direct experimental data, appropriate for materials whose constitutive response is difficult to calibrate from first principles.

3 Materials and Methods

3.1 Design Parameterization. We parameterize a family of tensegrity-inspired unit cells via four groups of design variables:

- **PLA compression members.** Strut diameter d_s and length ℓ_s, with the slenderness ratio ℓ_s/d_s bounded away from buckling-dominated regimes.
- **TPU tension elements.** Cross-sectional area A_t (or equivalent diameter d_t for circular cross-sections) and pre-tension geometry imposed during printing.
- **Connectivity topology.** Selected from a discrete set of candidate unit cells (e.g., truncated-tetrahedral, prism-based, and octahedral variants); the design vector encodes a categorical choice.
- **Unit-cell tiling.** Number of cells $N_x \times N_y \times N_z$ and the orientation of each cell relative to the impact direction.

Table 1 ***Placeholder.*** **Design variables, units, lower/upper bounds, manufacturing feasibility constraints, and categorical encoding.**

Column A	Column B	Column C
	(table contents pending)	

3.2 Multi-Material Fabrication. Specimens are fabricated on a multi-material FDM printer using PLA for the rigid struts and TPU for the soft tension elements. Following ~~**(author?)** [12]~~Ye et al. [12], the design uses a core-wrapping strategy in which each PLA strut is encapsulated by continuous TPU skin to provide robust mechanical interlocking and prevent interface delamination during cyclic loading.

Figure placeholder: Fabrication workflow: parameter → CAD → slicer → multi-material print → post-process and inspect.

Fig. 2 ***Placeholder.*** **Fabrication workflow: parameter → CAD → slicer → multi-material print → post-process and inspect.**

3.3 Experimental Characterization. Each printed specimen is subjected to two physical tests:

3.3.1 Quasi-static compression.. A uniaxial load–displacement curve is recorded on a benchtop electromechanical load frame at a controlled crosshead rate. Reported metrics are

$$\mathrm{SEA} = \frac{1}{m}\int_0^{\delta_d} F(\delta)\,\mathrm{d}\delta, \qquad \eta_c = \frac{\int_0^{\delta_d} F(\delta)\,\mathrm{d}\delta}{F_{\max}\,\delta_d}, \tag{1}$$

where m is the specimen mass, δ_d is the densification displacement, $F_{\max}$ is the peak transmitted force, and η_c is the compaction efficiency.

3.3.2 Drop-weight impact.. Following ~~**(author?)** [11]~~Pajunen et al. [11], specimens are subjected to drop-weight impact tests with instrumented load cells; we report the peak transmitted force and the energy-absorption plateau characteristics. Slip resistance and traction at the specimen–floor interface—governed by the tip geometry rather than the internal lattice—follow the test methodology of ISO 11334-4 [43] for walking-aid tips and are out of scope for the present axial-impact study.

3.4 Experiment-Driven Bayesian Optimization Loop. The optimization loop maintains a Gaussian-process (GP) surrogate over the design space defined in Section 3.1. After each round of physical experiments, the GP is updated and an acquisition function is maximized to recommend the next batch of designs to fabricate. Implementation uses BoTorch [21] with the log-Expected-Improvement (LogEI) acquisition [36] for the single-objective baseline and noisy Expected Hypervolume Improvement (qNEHVI) [37] for the multi-objective case (simultaneously maximizing SEA and compaction efficiency while constraining peak transmitted force). When experimental noise dominates (e.g., between-print variability), we additionally evaluate the input-noise-robust acquisition of ~~**(author?)** [38]~~ Daulton et al. [38] and the evolution-guided constrained variant of ~~**(author?)** [39].~~Low et al. [39]. The discrete connectivity-topology variable is handled with the mixed-search-space construction of [41]; the BOCS combinatorial baseline of [40] is reported for comparison.

Figure placeholder: Closed-loop BO schematic: design proposal → multi-material print → compression and drop-weight tests → GP update and acquisition maximization → next proposal.

Fig. 3 ***Placeholder.*** **Closed-loop BO schematic: design proposal → multi-material print → compression and drop-weight tests → GP update and acquisition maximization → next proposal.**

4 Results

4.1 Convergence of the BO Loop.

4.2 Pareto-Optimal Designs.

4.3 Reproducibility Across Print Replicates.

Figure placeholder: Best-so-far SEA (and Pareto hypervolume for the multi-objective runs) vs. number of physical experiments, averaged over random seeds; baselines: LHS and random search.

Fig. 4 ***Placeholder.*** **Best-so-far SEA (and Pareto hypervolume for the multi-objective runs) vs. number of physical experiments, averaged over random seeds; baselines: LHS and random search.**

Figure placeholder: Pareto front in (peak transmitted force, SEA) space; representative geometries annotated.

Fig. 5 ***Placeholder.*** **Pareto front in (peak transmitted force, SEA) space; representative geometries annotated.**

5 Discussion

Topics to address:

- Comparison to the simulation-driven multifidelity framework of ~~**(author?)** [17]~~ Mo et al. [17] and the single-condition fabrication study of ~~**(author?)** [11]~~Pajunen et al. [11].
- Practical considerations: TPU–PLA interface durability under cyclic loading; sensitivity of optimal designs to print parameters (extrusion temperature, layer height); transferability to TPU/PETG or other high-temperature thermoplastics.
- Path forward toward integrated tread+lattice structures, miniaturization to crutch-tip envelopes (from the Edison crutch-tip synthesis), and field testing. Long-term crutch use is associated with substantial upper-extremity overuse injury and ground-reaction-force impulses [44–48]; existing spring-loaded [49,50] and polymer-damped [51] tip designs reduce loading rate but not peak transmitted force, leaving an open design opportunity that an architected tensegrity-inspired insert is well-positioned to address. Recent open-source 3D-printed crutch work [52] and expert evaluations of orthopaedic caps [53] suggest a plausible translation path; adjacent footwear and orthotic lattice studies [54–56] provide additional transferable design heuristics.

5.0.1 Limitations..

6 Conclusions

We have presented an experiment-driven Bayesian-optimization framework for designing multi-material 3D-printed tensegrity-inspired energy absorbers using PLA struts and TPU tension elements. The framework parameterizes a family of unit cells, fabricates each candidate on a single multi-material FDM platform using a core-wrapping strategy [12], characterizes the response under quasi-static compression and drop-weight impact, and updates a Gaussian-process surrogate [21] that recommends the next batch of designs. The approach is broadly applicable to architected absorbers whose constitutive response is difficult to calibrate from first principles and where physical experiments—rather than simulations—are the natural ground truth.

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Conflict of Interest

The authors declare no conflict of interest.

Table 2 ***Placeholder.*** **Mean and CV of SEA, peak force, and compaction efficiency across *n* replicate prints for the top-*k* designs.**

Column A	Column B	Column C
(table contents pending)		

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